



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SEMESTER 1

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SECJ3553-02 ARTIFICIAL INTELLIGENCE (AI)

ASSIGNMENT 2

MADE BY:

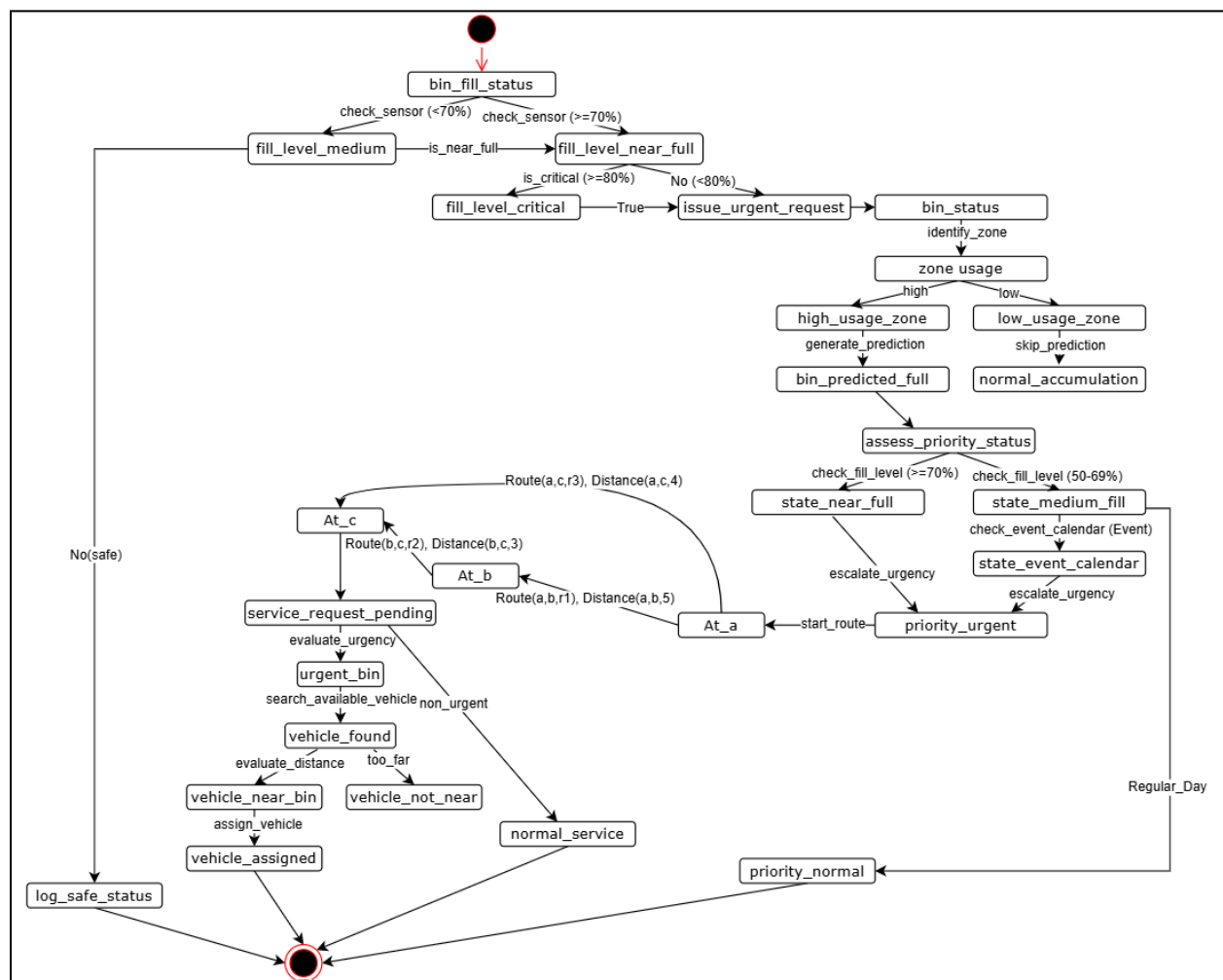
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Introduction

This assignment transitions from defining Knowledge Representations (KR) to formally modeling the decision making process of the Smart Waste Management System using a directed State Space Search for all KRs. This AI is essential as it systematically defines the system's states and actions to determine an optimal response to changing conditions. We utilize this framework to analyze all five Knowledge Representation by defining the Initial State, Actions, Goal State, and Path Cost, establishing a logical, traceable, and executable path for the AI to identify near-full bins and issue an urgent collection request, thereby supporting the project's goal of maintaining cleanliness and operational efficiency.

1.0 State Space Search



2.0 Explanation of States and Action

State (Nodes): Represents the current status

State	Description
bin_fill_status	The starting state where the system receives real time sensor data from the bin.
fill_level_medium	State where the fill level is below the NearFull threshold (< 70%)
fill_level_near_full	State where the fill level is $\geq 70\%$, corresponding to the NearFull(x) predicate from KR #1.
fill_level_critical	State where the fill level is $\geq 80\%$, requiring immediate collection.
issue_urgent_request	Goal State: The state where the AI signals the need for collection planning.
log_safe_status	Nothing happens where the system logs the data and returns to monitoring.
bin_status	The starting state where the system initiates the monitoring process for a specific bin.
zone_usage	State where waste generation rate of the bins is identified.
high_usage_zone	State where the bin is located in a zone that produces waste quickly.
low_usage_zone	State where the bin is located in a zone with normal waste generation.
bin_predicted_full	Goal state: The state where system predicts the bin will be full at future time t, based on high usage zone patterns, matching the PredictFull(x,t) predicate from KR #2.
normal_accumulation	End state: The state where the bin is expected to fill at a normal rate, no early prediction is generated, the bin returns to be monitored.
assess_priority_status	The starting state where the system initiates the priority evaluation process for a specific bin.
state_near_full	State where the bin's fill level is confirmed to be $\geq 70\%$, matching the NearFull(x) predicate.

state_medium_full	State where the bin's fill level is between 50% and 69%, requiring further context checks.
state_event_day	State where the system confirms that today is a special event day, implying higher waste generation.
state_regular_day	State where the system confirms today is not an event day.
priority_urgent	Goal State: The state where the bin is marked as "Urgent", triggering an immediate collection request.
priority_normal	End Path: The state where the bin is marked as standard priority, and no immediate alert is needed.
At_a	The lorry is currently at location a
At_b	The lorry is currently at location b
At_c	The lorry is currently at location c where Bin(c) is true
service_request_pending	State where a bin requires evaluation for collection.
urgent_bin	State where the bin is confirmed to require urgent service.
non_urgent_bin	State where the bin does not require urgent service.
vehicle_found	State where at least one vehicle is available.
vehicle_near_bin	State where an available vehicle is near the bin.
vehicle_not_near	State where no available vehicle is assigned to the bin.
vehicle_assigned	Goal State: The state where a vehicle is assigned to the bin.
normal_service	End State: The state where the bin is marked as standard service and no immediate service is needed.

Actions (Edges): Represent the AI's checks and status transitions.

Action	Description
check_sensor	Action to initiate reading and updating the current fill level.
is_near_full	Action to check if the level is $\geq 70\%$ (the threshold(t)).
is_critical	Action to check if the level is $\geq 80\%$ (a secondary, higher threshold).
Yes/True	Action for a condition check resulting in true.

No/False	Action for a condition check resulting in false.
identify_zone	Action to identify which zone the bin belongs to.
generate_prediction	Action to generate and predict early fullness.
skip_prediction	Action to log the bin as safe/non-urgent and continue normal monitoring.
high	Action for condition check resulting in the bin at a high usage zone.
low	Action for condition check resulting in the bin at a low usage zone.
check_fill_level	Action to read sensor data and categorize the bin as Low, Medium, or Near Full.
check_event_calendar	Action to query the system's database to verify if the current date is marked as EventDay(d).
escalate_urgency	Action to flag the bin as urgent based on the logic rules provided in KR #3.
maintain_normal	Actions to log the bin as safe/non-urgent and return to monitoring.
Route(a,b,r1), Distance(a,b,5)	A route r1 exists connecting location a to b. The distance between location a to b is 5.
Route(b,c,r2), Distance(b,c,3)	A route r2 exists connecting location b to c. The distance between location b to c is 3.
Route(a,c,r3), Distance(a,c,4)	A route r3 exists connecting location a to c. The distance between location a to c is 4.
evaluate_urgency	Action to determine if the bin is urgent to collect.
search_available_vehicle	Action to search for available vehicles.
evaluate_distance	Action to check distance between vehicle and bin.
assign_vehicle	Action to assign vehicle to bin
find_nearest	Action to find the nearest vehicle to the bin.
maintain_normal	Actions to log the bin as safe/non-urgent and return to monitoring.

3.0 Problem Formulation and Explanation

The combined state space problem is formulated as a single sequential decision process that enables the AI system to move from environmental sensing to operational execution.

Initial State

- **bin_fill_status**
The system begins by monitoring bin fill levels using real-time sensor data.

Actions and State Transitions

Let $S(x)$ be the set of action–state pairs that define valid transitions:

- $S(\text{bin_fill_status}) \rightarrow \{\text{fill_level_medium}, \text{fill_level_near_full}\}$
The system categorizes the bin based on a predefined threshold (70%).
- $S(\text{fill_level_near_full}) \rightarrow \{\text{issue_urgent_request}\}$
Near-full bins trigger further analysis.
- $S(\text{issue_urgent_request}) \rightarrow \{\text{zone_usage}\}$
The system evaluates waste accumulation patterns based on zone type.
- $S(\text{zone_usage}) \rightarrow \{\text{high_usage_zone}, \text{low_usage_zone}\}$
High-usage zones lead to early overflow prediction.
- $S(\text{high_usage_zone}) \rightarrow \{\text{bin_predicted_full}\}$
The system predicts future bin fullness.
- $S(\text{bin_predicted_full}) \rightarrow \{\text{assess_priority_status}\}$
Priority evaluation is initiated.
- $S(\text{assess_priority_status}) \rightarrow \{\text{priority_urgent}, \text{priority_normal}\}$
The system determines urgency based on fill level and context.
- $S(\text{priority_urgent}) \rightarrow \{\text{At}(a) \rightarrow \text{At}(c)\}$
Route optimization is performed to reach the bin via the shortest path.
- $S(\text{At}(c)) \rightarrow \{\text{service_request_pending} \rightarrow \text{vehicle_found} \rightarrow \text{vehicle_assigned}\}$
Vehicle availability is checked and the nearest vehicle is assigned.

Goal State

- **vehicle_assigned**
The system successfully assigns a collection vehicle to an urgent bin.

Path Cost

- The path cost represents the number of decision steps and operational checks required to reach the goal state. Urgent bins incur higher path costs due to additional reasoning steps (prediction, prioritization, routing and assignment), while safe bins terminate early with minimal cost.

4.0 Supporting the Proposed KR

The combined State Space Search operationalizes all five Knowledge Representations into a single executable reasoning framework:

1. KR #1 (Bin Fill-Level Knowledge)

Implemented through transitions from `bin_fill_status` to `fill_level_safe` or `fill_level_near_full`.

2. KR #2 (Waste Accumulation Prediction Knowledge)

Represented by `zone_usage`, `high_usage_zone` and `bin_predicted_full`, enabling early overflow prediction.

3. KR #3 (Collection Priority Knowledge)

Modeled through `assess_priority_status`, `priority_urgent` and `priority_normal`, integrating contextual urgency rules.

4. KR #4 (Route Optimization Knowledge)

Implemented using location states `At(a)` and `At(c)`, where the system selects the shortest route.

5. KR #5 (Vehicle Assignment Knowledge)

Realized through `service_request_pending`, `vehicle_found` and `vehicle_assigned`, ensuring the nearest available vehicle is dispatched.

By integrating all five KRs into a unified state space, the system demonstrates how logical knowledge representations are transformed into a coherent, step-by-step AI decision process. This combined formulation ensures timely waste collection, efficient resource utilization and effective operational decision-making, fully supporting the objectives of the Smart Waste Management System.